

BERT-Based Fine-Tuning for Automated Tagging of Robbery Crime Narratives

Lenin G. Falconi¹[0000–0003–4402–6643],
Myriam Hernandez-Alvarez¹[0000–0003–4718–0400], and
Ángel Leonardo Valdivieso Caraguay¹[0000–0002–3502–020X]

Escuela Politécnica Nacional, Quito 170525, Ecuador
<https://www.epn.edu.ec>
`{lenin.falconi,myriam.hernandez,angel.valdivieso}@epn.edu.ec`

Abstract. Accurate classification of crime narratives is vital for reliable public safety statistics. In Ecuador, the Comisión Especial de Estadística de Seguridad, Justicia, Crimen y Transparencia (CEESJCT) manually categorizes robbery incident reports, which is a time-consuming process. While transformer-based models have shown success in natural language processing tasks, their application to Ecuadorian legal and security texts in the Spanish language remains underexplored. This study addresses this gap by developing an automated classification system using a BERT model tailored to Spanish robbery narratives. Utilizing transfer learning and subsequent fine-tuning on an expanded, labeled dataset, the system significantly improves classification performance. Initial transfer learning achieved moderate accuracy (80.5%) but faced difficulties with semantically similar categories. Fine-tuning notably increased minority-class recall (up to 30%) with an improved accuracy (90.3%). The final implementation, which increased the number of categories to 11, achieved 95.5% accuracy with robust and consistent results on both police and judicial narratives. Collaboration with Ecuadorian institutions, including the Fiscalía General del Estado (FGE) and Instituto Nacional de Estadística y Censos (INEC), ensures model credibility.

Keywords: Legal Natural Language Processing · NLP · Natural Language Processing · transformers · fine-tuning · transfer learning.

1 Introduction

Deep learning models (DL) have yielded remarkably successful outcomes across a wide range of applications (e.g., image classification, object detection, natural language processing)[17]. In various text classification tasks, such as sentiment analysis, news categorization, question-answering, and natural language inference, DL has outperformed traditional *Machine Learning (ML)* methods[14], a testament primarily to its inherent generalization and robustness[3]. However, depending on the problem to be solved, designing a DL architecture demands a substantial volume of data to attain the desired generalization performance. For

example, in image processing and computer vision, datasets such as ImageNet (14,197,122) and Microsoft COCO (2.5 million) provide extensive data per category [17], while in *Natural Language Processing (NLP)*, resources like WebTex, composed of millions of web pages, have been pivotal in training models such as GPT-2 [18].

Although the success of DL hinges on large datasets, specialized hardware [18][16], and model architecture, for niche domains with limited data, *Transfer Learning (TL)* and *Fine-Tuning (FT)* enable knowledge transfer from source to target domains by reusing early-layer features and adapting task-specific layers [28, 9]. The legal documentation domain is a promising field for applying NLP, where digital information can be leveraged to develop tools that optimize legal workflows. However, applications face language limitations, as LexGlue, the most widely used benchmark, focuses on English despite the growing interest in judicial outcome prediction [15, 11, 26], legal article prediction from text, and document classification [27, 6]. While transformer models like mBERT enable cross-lingual transfer, legal terminologies and system differences necessitate domain-specific pretraining (e.g., LegalBERT [5]) or hybrid approaches [13], particularly for low-resource languages [20].

In Ecuador’s penal code (COIP), robbery is defined in Article 189. However, for public security analytics, this is further classified into specific subtypes including, but not limited to: home robbery, street robbery, business robbery, vehicle parts and accessories theft, car theft, and motorcycle theft. These classifications are manually identified by the CEESJCT using *crime incident reports*. To automate this, we propose a transformer-based classifier that uses multilingual DistilBERT [19]. We implemented a three-phase approach: **Phase 1: Classification with Transfer Learning.** The transformer model is used as a feature extractor, followed by fully connected layers and dropout, achieving 80.5% accuracy (F1: 0.80). Even though the model’s bidirectional attention mechanism and subword tokenization (WordPiece) proved essential for capturing semantic relationships in crime narratives (particularly when some words remain misspelled and uncorrected for legal reasons protecting the victim’s statement), it struggled with minority classes. **Phase 2: Fine-Tuning for Contextual Adaptation.** Using a learning rate of 3×10^{-6} improved accuracy to 90.3% (F1: 0.90) by resolving semantic ambiguities and achieving significant gains for less frequent categories. Stages 1 and 2 used a dataset of 431,669 robbery reports (2014–2022), tokenized into sequences of up to 300 words, and split into training (63%), validation (16%), and test (21%) sets. **Phase 3: Scaling to Complex Taxonomies.** The model is fine-tuned to predict a total of 11 categories, extending the original 6. A merged, validated dataset of 1.1 million narratives was developed by combining police reports with criminal complaints filed in the prosecutor’s office. The model was trained using a Tensor Processing Unit (TPU) to process the entire dataset. A notable improvement in accuracy performance was achieved (95.5% accuracy with an F1-score of 0.95).

This research makes three significant contributions to Legal Natural Language Processing (LNLP) for Spanish-language documents. First, we introduce

a novel, large-scale dataset comprising over one million annotated robbery narratives from Ecuadorian robbery reports, addressing a critical gap in Spanish-language legal text resources. Second, we develop an effective three-phase fine-tuning approach that adapts a pre-trained BERT model to the Ecuadorian legal context, achieving 95.5% accuracy while expanding its capability to classify a total of 11 categories. Third, our model has been successfully deployed operationally by Ecuador’s FGE in their Statistics and Information Systems Directorate ¹, automating their previously manual classification process. To our knowledge, this represents the first successful application of Transformer-based models for analyzing crime narratives in Ecuadorian Spanish, demonstrating both the technical feasibility and practical value of advanced NLP techniques for legal documentation challenges in Spanish-speaking contexts while establishing a foundation for future Latin American legal NLP applications.

The remainder of this article is structured as follows: Section 2 reviews concepts about LNLP. Related works are presented in Section 3. The proposed methodology, including dataset generation, is detailed in Section 4. Section 5 presents the experimental results. Finally, Section 6 discusses the main findings, conclusions, and directions for future work.

2 Legal Natural Language Processing

In this research, we propose a transformer-based model for robbery offense classification aligned with categorical frameworks established by both the Prosecutor’s Office and INEC. Legal documents present unique challenges compared to general-domain texts, characterized by three distinctive features: (1) inherent structural complexity, (2) substantial document length, and (3) specialized juridical terminology. These characteristics have led to the emergence of LNLP as a specialized NLP subdomain [4, 29].

Within this paradigm, Legal Text Classification (LTC) refers to the task of categorizing legal documents into predefined juridical categories. While LTC shares fundamental principles with general TC, it introduces specific technical challenges including, but not limited to: high class cardinality (typically ranging from dozens to hundreds of categories), multi-label classification requirements, and the need for domain-specific feature engineering. The complexity of LTC increases exponentially with the number of potential legal categories and their hierarchical relationships within juridical systems.

NLP for legal documents primarily focuses on two categories of approaches: *Embedding Methods* and *Symbol-Based Methods* [29]. Embedding Methods, such as transformer-based models (e.g., BERT), derive high-level representations from large corpora and offer strong predictive performance, but their opacity raises concerns about interpretability. Conversely, Symbol-Based Methods explicitly model legal knowledge and facilitate reasoning over structured representations, thereby providing greater transparency, though often at the expense of accuracy.

¹ Dirección de Estadística y Sistemas de Información

Recent work in explainable AI seeks to address this limitation in embedding methods. For example, attention-based techniques such as *attention rollout* and *attention flow* aggregate attention weights across layers in transformers, offering insight into model decisions and revealing a degree of inherent explainability [1].

The Transformer architecture, introduced in [24], represents a significant advancement in NLP by incorporating a self-attention mechanism that allows DL models to effectively capture long-range dependencies and facilitates efficient parallel training. This innovation supplanted the previous state-of-the-art recurrent neural network (RNN) approaches, including Long Short-Term Memory (LSTM) networks [23]. Transformers employ an encoder-decoder architecture augmented by self-attention and transfer learning (TL) methodologies.

BERT [7], a notable application of the Transformer encoder, generates deeply bidirectional contextual representations through an exclusively encoder-based architecture. Its pre-training utilizes self-supervised objectives such as masked language modeling (MLM) and next sentence prediction (NSP), enabling the model to acquire rich contextual information from extensive corpora.

3 Related Works

This section reviews related work in LTC research, focusing on three critical dimensions: (1) architectural innovations, (2) dataset characteristics and domain adaptation strategies, and (3) comparative performance. Table 1 provides a comparison of these approaches. While existing research has predominantly focused on English-language legal corpora our work represents, to our knowledge, innovative in exploring transformer architectures in Spanish legal documents attached to Ecuadorian reality.

In their research work, [21] tackle large-scale multi-label legal text classification using transformer architectures (BERT, RoBERTa, DistilBERT, XLNet, M-BERT). Through systematic evaluation on the JRC-Acquis (multilingual) and EURLEX57K (English) datasets, they demonstrate that combining gradual layer unfreezing, discriminative learning rates, and domain-specific pretraining achieves state-of-the-art performance, surpassing LSTM baselines by significant margins. The authors further propose standardized dataset splits to facilitate reproducible research, establishing transformers as the new benchmark for legal document classification with complex taxonomies like EuroVoc.

Zero-shot cross-lingual transfer with multilingual transformers such as M-BERT and M-DistilBERT is explored in [22]. By fine-tuning the models solely on English EURLEX57K documents and subsequently evaluating them on French and German translations, target-language performance is achieved that closely matches models trained on all three languages. Notable advancements comprise continued pretraining on legal corpora and the use of progressive unfreezing during fine-tuning, which together underscore the capacity of transformer architectures to bridge linguistic gaps, particularly in low-resource legal NLP contexts.

The cross-linguistic applicability of transformers is further validated by [2] for the classification of Turkish legal texts. Their comparative analysis reveals

that transformer-based models outperform traditional ML methods (e.g., SVM, logistic regression) in accuracy, despite limited training data. While specific classification categories remain unspecified, this study crucially demonstrates that transformers’ superiority extends beyond English to other languages.

Addressing the challenge of document length, [25] systematically evaluates BERT adaptations for U.S. Supreme Court decisions from the Supreme Court Database (SCDB). They find that domain-adapted Legal-LongFormer and Legal-BERT outperform their general-domain counterparts. This underscores the significance of domain-specific pretraining and optimized strategies for handling lengthy documents in legal text classification [21].

Table 1. Comparative analysis of legal text classification approaches

Ref	Model	Task	Dataset	Language	Key Results
[21]	BERT, RoBERTa, DistilBERT, XLNet, M-BERT	Multi-label classification	JRC-Acquis, EURLEX57K	Multilingual, English	0.661 (F1) on JRC-Acquis
[22]	M-BERT, M-DistilBERT	Cross-lingual transfer	EURLEX57K Extended	Models pre-trained in Multilingual. FT in English	34% improvement on French, 87% improvement on German
[2]	Transformer-based	Document classification	Turkish Legal Corpus	Turkish	DL models improve over traditional ML
[25]	BERT, RoBERTa, Legal-BERT, LongFormer, Legal-LongFormer	Long document classification	SCDB	English	80.1% accuracy (15 categories), 60.9% accuracy (279 categories) with Legal-BERT
This work	DistilBERT	Multi-label classification	CEESJCT robbery reports	Spanish	95.5% accuracy (11 categories)

3.1 Discussion and Findings

The reviewed literature indicates that BERT models are widely utilized for LTC. Transformer-based architectures consistently outperform traditional ML methods in this domain [2]. Fine-tuning pre-trained BERT weights to the target task is a standard approach. Also, our methodology is comparable to that of [25], wherein models are trained for both broader and fine-grained legal categories. A recurrent challenge with BERT-based approaches concerns the processing of lengthy documents, due to BERT’s tokenization limits.

Distinct from previous works, our study specifically addresses the classification of robbery types. Most literature focuses on broader legal categories, and while some studies may implicitly include robbery within categories such as "criminal law" or "property offenses", explicit attention to robbery classification is absent, marking our focus as potentially novel and expanding the exploration in the application of transformer-based models in the legal domain.

Another important difference in this work is related to the nature of its data. While it is recognized that LNLP has unique characteristics compared to general NLP, the narratives in our dataset originate from three principal sources: citizen crime reports, police accounts, and prosecutor’s office reports. Consequently, our dataset encompasses a blend of technical texts and non-technical victim-generated descriptions. These factors substantiate the application of BERT-based models for the current text classification task.

The implementation of the model at FGE proves relevant not only due to the automation of the classification process but also by demonstrating the feasibility of adopting state-of-the-art AI and NLP in the legal domain. This approach offers the potential to generate solutions and foster scientific advancement. Collaborating with governmental entities is essential for achieving objectives that further the effective application of AI.

4 Methodology

This section details the development of a transformer-based robbery classification model for Ecuadorian crime reports, structured around three pillars: dataset preparation, training protocol, and performance evaluation. The model was developed in three stages: initial transfer learning (TL) on a reduced dataset of 6 categories ($D_{\kappa=6}$), followed by fine-tuning (FT) on the same dataset, and concluding with FT on an expanded dataset of 11 categories ($D_{\kappa=11}$).

4.1 Dataset Generation

Basic Taxonomy ($D_{\kappa=6}$) Dataset To train the model, an initial target dataset $D_{\kappa=6,N}^T$ with 6 categories was generated by updating an SQL table of robbery records from CEESJCT (as of June 8, 2022), obtaining 671,708 records. Next, the crime narratives were extracted and paired with the previous table using the record’s key, yielding 671,146 records. Subsequently, we cleaned the text sequences by removing non-alphanumeric characters and converting text to lowercase by default (though distilbert-base-multilingual-cased preserves case sensitivity). The statistical analysis of word counts l_w of each document X_i from the gathered robbery narratives gathered showed a mean (μ) = 98.34, a standard deviation of (δ) = 77.38, showing high variability in narrative lengths. The histogram of the word counts presented a bimodal distribution, with peaks at $l_w \in \{7, 100\}$. Also, $\max(l_w) = 914$ words. Quartile analysis showed a median of 52 words per X_i , with $q_1 = 33$, $q_3 = 137$, and an upper threshold $l_{w_{sup}} = 293$ words. Given that the DistilBERT model has a maximum sequence length of 512 tokens and produces 768-dimensional embeddings [19], we constructed dataset $D_{\kappa=6,N}^T$ consisting of narratives where the number of words of each sample X_i satisfied $35 < l_{w_i}(X_i) \leq 300$. This selection resulted in a total of $N = 671,708$ records. The dataset $D_{\kappa=6,N}^T$ was partitioned into training ($D_{\kappa=6,train}^T$), validation ($D_{\kappa=6,valid}^T$), and testing ($D_{\kappa=6,test}^T$) subsets to facilitate model training and evaluation. The training set comprises 273,336 records (63.32% of the dataset),

while the validation and test sets contain 68,333 (15.83%) and 90,000 (20.85%) records, respectively. The validation set was derived by splitting the training data in an 80:20 ratio.

Extended Taxonomy ($D_{\kappa=11}$) Dataset We constructed an extended target dataset comprising 11 distinct categories by integrating police reports with corresponding prosecutor’s office records. This integration significantly expanded our training corpus, and enhanced the model’s exposure to diverse textual patterns. The final dataset contained 1,109,335 records, with the distribution across categories detailed in Table 2. The dataset creation process involved the following steps:

1. **Initial Data Extraction:** Crime reports from 2014–2022 were extracted and filtered for robbery cases from CEESJCT, yielding to 735,045 records.
2. **Narrative Retrieval:** Crime narratives were obtained for each of the previous records, totaling 725,079. However, non-robbery cases were excluded if found, which produced a total of 723,435 records.
3. **Preprocessing:** Text was preprocessed through lowercasing and removal of non-alphanumeric characters.
4. **Dataset Integration:** Police and prosecutor narratives were combined into a single dataset, yielding to 1,446,870 records.
5. **Quality Control:** A statistical analysis over the text was carried out. Narratives with 50–400 words were retained (upper fence: 389.5 words), reducing the dataset to 1,140,728 records. Crime labels were also standardized (e.g., removing accents).
6. **Category Refinement:** Non-compliant labels (48 categories) were discarded with the exception of the *OTHER ROBBERIES* category, which was retained. The final dataset, with 1,109,335 records, was randomly shuffled.
7. **Dataset Split:** The dataset was partitioned into training (807,468 samples $\approx 72\%$), validation (201,867 samples $\approx 18\%$), and test (100,000 samples $\approx 10\%$) sets.

4.2 Model Training

Problem Formulation In this study, we address the task of classifying crime report narratives of robbery into k predefined categories. A language model estimates the conditional probability of the next token given a preceding sequence of tokens. Within this framework, TC can be formalized as in (1): given a sequence of words $\mathbf{x}$, the goal is to predict its corresponding label $\hat{y}$. Training a TC model thus involves learning a function f that maps a target dataset D^T , consisting of text sequences $X_i \in D = \{X_1, X_2, \dots, X_N\}$ each formed by s sentences and w_s words², to one of k categories, as shown in (2).

² We adopt the notation presented in [12].

Table 2. Extended Taxonomy Dataset

Basic Taxonomy	Extended Labels	Total
HOME ROBBERY	HOME ROBBERY	172,264
STREET ROBBERY	STREET ROBBERY	421,497
BUSINESS ROBBERY	BUSINESS ROBBERY	74,088
VEHICLE PARTS AND ACCES- SORIES THEFT	VEHICLE PARTS AND ACCES- SORIES THEFT	154,546
CAR THEFT	CAR THEFT	90,038
MOTORCYCLE THEFT	MOTORCYCLE THEFT	119,128
NO INFORMATION	OTHER ROBBERIES	43,468
	WATER VESSEL ROBBERY	9,407
	SOCIAL ORGANIZATION ROBBERY	3,087
	EDUCATIONAL INSTITUTION ROB- BERY	17,252
	PUBLIC INSTITUTION ROBBERY	4,560
Total		1,109,335

For this work, we selected the distilbert-base-multilingual-casedmodel [19], a distilled version of BERT, which retains 97% of BERT’s language understanding capabilities while significantly reducing the number of parameters. Moreover, it supports multilingual tasks, including Spanish, aligning well with the linguistic requirements of our dataset.

$$\hat{y} = \underset{y \in \{1, \dots, k\}}{\operatorname{argmax}} P(y \mid w_1, w_2, \dots, w_i) \quad (1)$$

$$f : D \longrightarrow \mathbb{Z}^k \quad (2)$$

To enable processing by a transformer-based language model, textual sequences must first undergo tokenization and be transformed into numerical representations. This involves segmenting the text into discrete tokens and subsequently mapping these tokens to vector embeddings. Let $\text{Tokenize} : \mathcal{X} \rightarrow \mathbb{Z}^\lambda$ be the tokenization function that maps text sequences to token IDs, where $\mathcal{X}$ is the text space and λ is the maximum sequence length. For each input sequence $X_i \in D_N^T$, the tokenizer produces:

$$(\boldsymbol{\nu}_i, \boldsymbol{\tau}_i) = \text{Tokenize}(X_i) \quad (3)$$

where:

- $\boldsymbol{\nu}_i \in \mathbb{Z}^\lambda$ are the token IDs
- $\boldsymbol{\tau}_i \in \{0, 1\}^\lambda$ is the attention mask
- $\lambda = 512$ for distilbert-base-multilingual-cased

The token IDs are passed through the embedding layer, which maps each token to a $d = 768$ dimensional vector representation. The embedding matrix $W_{\text{embedding}} \in \mathbb{R}^{119,547 \times 768}$ uses a $V = 119,547$ vocabulary, where each row corresponds to a unique token, thereby providing dense vector representations.

$$\mathbf{E} = \text{Embedding}(\boldsymbol{\nu}_i) \in \mathbb{R}^{\lambda \times d} \quad (4)$$

The encoded numerical representations of the text are then processed by distilbert-base-multilingual-cased:

$$\mathbf{H} = \text{BERT}(\mathbf{E}, \tau_i) \in \mathbb{R}^{\lambda \times d} \quad (5)$$

The classification layer comprises a feedforward layer that maps the hidden state vector output from the encoder (or CLS embedding) $\mathbf{h}_{[\text{CLS}]}$ to the k categories:

$$\hat{\mathbf{y}}_i = \text{softmax}(\mathbf{W}\mathbf{h}_{[\text{CLS}]} + \mathbf{b}) \in \mathbb{R}^k \quad (6)$$

where:

- $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^d$ is the input feature vector of $d = 768$
- $\mathbf{W} \in \mathbb{R}^{k \times d}$ is the classification weight matrix, mapping from embedding space to k classes.
- $\mathbf{b} \in \mathbb{R}^k$ is the bias vector
- k is the number of target categories
- A softmax activation to obtain probabilities $p_i = \frac{\exp(y_i)}{\sum_{j=1}^k \exp(y_j)}$, where p_i is the probability of class i

In this work, we experiment with TL and FT to adapt the weights of a pre-trained model from a large source dataset D^S to the characteristics of a target dataset D^T . We adopt the TL definition provided by [8]:

Definition 1 (Transfer Learning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$, with a target task $\mathcal{T}^T$, the TL operation seeks to improve the learning of a prediction function $\phi^T(\cdot)$ in $\mathcal{D}^T$ using the knowledge acquired in $\phi^S(\cdot)$ in $\mathcal{T}^S$, through ϕ^S ; where $\mathcal{D}^S \neq \mathcal{D}^T$ or $\mathcal{T}^S \neq \mathcal{T}^T$.*

Considering the mentioned definition of TL, (7) represents the TL operation applied to the text classification problem, starting from a pre-trained natural language model; where L is the number of layers of the pre-trained model.

$$\mathbb{T}_L \langle \phi^S(\mathcal{D}^S), \mathcal{D}^T \rangle = \mathcal{A} \left(\phi^S(\mathcal{D}^T) \Big|_0^L \right) \quad (7)$$

Consequently, the prediction generation is represented by Equation (8)

$$\mathcal{Y}^T = \mathbb{T}_L \langle \phi^S(D^S), \mathcal{D}^T \rangle = \phi^T(D^T) \quad (8)$$

Similarly, we use the definition of FT as presented in [8]:

Definition 2 (Fine Tuning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$, with a target task $\mathcal{T}^T$, the FT procedure enhances the learning of a target function $\mathcal{Y}^T = \phi^T(\mathcal{I}^T)$ by retraining r layers of the original model $\mathcal{Y}^S = \phi^S(\mathcal{I}^S)$ with L layers and $\gamma = L - r$.*

The definition for FT is provided in (9), whereas the special case $\gamma = 0$, where $L = r$, is given in (10) and corresponds to full model tuning.

$$\mathbb{F}_{\mathbb{T}} \langle \phi^S(\mathcal{D}^S), \mathcal{D}^T \rangle = \mathcal{A} \left(\phi^S \left(\phi^S(\mathcal{D}^T) \Big|_0^{\gamma-1} \right) \Big|_{\gamma}^L \right) \quad (9)$$

$$\mathbb{F}_{\mathbb{T}} \langle \phi^S(\mathcal{D}^S), \mathcal{D}^T \rangle = \mathcal{A}(\phi^S(\mathcal{D}^T)) \quad (10)$$

Phase 1: Baseline Transfer Learning In this stage, TL is employed to adapt a pre-trained *distilbert-base-multilingual-cased* model $\phi^S(\cdot)$ for the target task $\phi^T(\cdot)$ by applying transformations $\mathcal{A}$ to its outputs, while keeping the original weights $\mathcal{W}^S = 134,735,616$ fixed. Input samples are encoded by the feature extractor, yielding tensors in $\mathbb{R}^{N \times 300 \times 768}$, where N represents the batch size. Global Max Pooling (GMP) condenses these to $\mathbb{R}^{N \times 768}$. The adaptation block, with $\mathcal{W}^T = 104,294$ trainable parameters, consists of Batch Normalization (BN), a fully connected layer of 512 units (FC_{512}), Dropout ($\text{Dro}_{0.1}$), two fully connected layers of 128 units each (FC_{128}), another Dropout layer, and an output layer of $\kappa = 6$ units (FC_{κ}). During training, only $\mathcal{W}^T$ is updated, as detailed in Equations (7), (8), and (11).

$$\mathcal{A} = \text{FC}_6 \circ \text{Dro}_{0.1} \circ \text{FC}_{128} \circ \text{Dro}_{0.1} \circ \text{FC}_{128} \circ \text{Dro}_{0.1} \circ \text{FC}_{512} \circ \text{BN} \circ \text{GMP} \quad (11)$$

Due to hardware constraints, a reduced subset of the data was employed in this phase: $d_{\text{train}}^T : 20,000 \subset D_{\text{train}}^T$, $d_{\text{valid}}^T : 4,000 \subset D_{\text{valid}}^T$, and $d_{\text{test}}^T : 4,000 \subset D_{\text{test}}^T$. The training parameters utilized in this stage are summarized in Table 3 Phase 1.

Phase 2: Fine-Tuning on Dataset $D_{\kappa=6}$ In this phase, we adopt FT for all the pre-trained model’s weights, as represented in (10), leveraging the inherent advantage that full FT does not necessitate supplementary output layers. As stated in [23], this procedure reduces loss and increases classification performance on the new task. Table 3 details the training hyperparameters utilized to FT the model in the basic taxonomy dataset in the Phase 2 column.

Phase 3: Scaling to Dataset $D_{\kappa=11}$ In this phase, we extend the scope of the FT model to the $D_{\kappa=11}$ dataset, which expands to a total of 11 categories. Procedures for building the aforementioned dataset were described in Section 4.1. The dataset was split into training, validation, and testing sets with 807,468, 201,867, and 100,000 samples, respectively. To FT the model using such a large dataset, we used a TPU v2[10]. The hyperparameters settings for this phase are presented in Phase 3 of Table 3.

Table 3. Training Parameters Across Phases

Parameter	Phase 1 (TL)	Phase 2 (FT)	Phase 3 (FT-extended)
Loss	Categorical Cross Entropy	Sparse Categorical Cross Entropy	Sparse Categorical Cross Entropy
Metrics	Categorical Accuracy	Sparse Categorical Accuracy	Sparse Categorical Accuracy
Optimizer	Adam	Adam	Adam
Learning rate	2×10^{-4}	3×10^{-6}	3×10^{-6}
Epochs	20	10	12
Early stopping	5 epochs	10 epochs	10 epochs
Batch size	16	16	128

Table 4. Phase 1 TL Classifier Performance on $D_{\kappa=6}$

class	precision	recall	f1-score	support
Car Theft	0.58	0.36	0.45	285
Motorcycle Theft	0.90	0.84	0.87	457
Vehicle Parts And Accessories Theft	0.79	0.79	0.79	613
Street Robbery	0.82	0.91	0.86	1,666
Business Robbery	0.82	0.70	0.75	328
Home Robbery	0.76	0.77	0.77	651
micro avg	0.80	0.80	0.80	4,000
macro avg	0.78	0.73	0.75	4,000
weighted avg	0.80	0.80	0.80	4,000

5 Experimental Results

Tables 4, 5, and 6 report the model’s performance across Phases 1, 2, and 3, respectively. Fine-tuning the model on the extended dataset with TPU acceleration led to improved performance and enabled the prediction of a broader range of robbery categories. The observed discrepancies in support counts between Phase 1 and Phase 2 are attributed to changes in category labeling introduced by CEESJCT. It is also evident that the distribution of samples across categories is imbalanced, reflecting the underlying social circumstances surrounding criminal activity. Consequently, we opted not to employ synthetic data augmentation for this study. The results further suggest that the *Public Institution Robbery* category should be merged into the *Other Robberies* category, due to the limited number of available samples.

6 Conclusions and Future Works

Transformer-based models have achieved state-of-the-art performance across a range of NLP tasks. In this work, we employed FT to address text classification for legal documents. The proposed methodology, based on FT, demonstrates that transformer architectures can be effectively adapted to the legal domain. Notably, our model demonstrates substantial robustness to class imbalance, with the exception of the *Public Institution Robbery* category. Due to the limited

Table 5. Phase 2 FT Classifier Performance on $D_{\kappa=6}$

class	precision	recall	f1-score	support
Business Robbery	0.78	0.79	0.78	297
Car Theft	0.95	0.88	0.92	363
Motorcycle Theft	0.95	0.93	0.94	392
Vehicle Parts & Accessories	0.85	0.92	0.88	663
Home Robbery	0.87	0.89	0.88	663
Street Robbery	0.94	0.92	0.93	1,671
Accuracy			0.90	4,000
Macro Avg	0.89	0.89	0.89	4,000
Weighted Avg	0.90	0.90	0.90	4,000

Table 6. Phase 3 FT Classifier Performance on $D_{\kappa=11}$

	precision	recall	f1-score	support
Educational Institution Robbery	0.925995	0.946599	0.936184	1,573
Motorcycle Theft	0.985211	0.988317	0.986762	10,785
Public Institution Robbery	0.724528	0.474074	0.573134	405
Car Theft	0.967669	0.980198	0.973893	7,878
Social Organization Robbery	0.750000	0.752688	0.751342	279
Street Robbery	0.969847	0.972298	0.971071	37,976
Business Robbery	0.878871	0.893877	0.886311	6,794
Home Robbery	0.944779	0.952335	0.948542	15,504
Vehicle Parts & Accessories Theft	0.977752	0.968340	0.973023	14,024
Water Vessel Robbery	0.965197	0.983452	0.974239	846
Other Robberies	0.817738	0.766006	0.791027	3,936
accuracy			0.954610	100,000
macro avg	0.900690	0.879835	0.887775	100,000
weighted avg	0.954050	0.954610	0.954175	100,000

number of instances associated with this particular type of robbery, it may be advisable to consolidate it within the broader *Other Robberies* category. The utilization of TPU hardware was pivotal for processing the expanded dataset, enabling both an increase in the number of target categories and overall model performance. These results highlight the practical potential of deploying ML solutions within legal institutions such as the FGE, where automated narrative classification may facilitate more efficient resource allocation and optimized operational workflows. Future research should prioritize the adaptation of novel NLP tasks, as addressed by transformer-based models, to LNLP. Such tasks encompass, but are not limited to, legal question answering, legal named entity recognition, and legal judgment prediction. The latter, in particular, necessitates an increased emphasis on model explainability, given the importance of interpretability for legal stakeholders.

Acknowledgments. The authors would like to thank Fiscalía General del Estado and INEC for their cooperation in the development of this research.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Abnar, S., Zuidema, W.: Quantifying attention flow in transformers. Proceedings of the Annual Meeting of the Association for Computational Linguistics pp. 4190–4197 (5 2020). <https://doi.org/10.18653/v1/2020.acl-main.385>, <https://arxiv.org/abs/2005.00928v2>
2. Akca, O., Bayrak, G., Issifu, A.M., Ganiz, M.C.: Traditional machine learning and deep learning-based text classification for turkish law documents using transformers and domain adaptation. 16th International Conference on INnovations in Intelligent SysTems and Applications, INISTA 2022 (2022). <https://doi.org/10.1109/INISTA55318.2022.9894051>
3. Alom, M.Z., Taha, T.M., Yakopcic, C., Westberg, S., Sidike, P., Nasrin, M.S., Esssen, B.C.V., Awwal, A.A.S., Asari, V.K.: The history began from alexnet: A comprehensive survey on deep learning approaches. CoRR **abs/1803.01164** (2018), <http://arxiv.org/abs/1803.01164>
4. Ariai, F., Demartini, G.: Natural language processing for the legal domain: A survey of tasks, datasets, models, and challenges. ACM Computing Surveys **1** (10 2024). <https://doi.org/10.1145/nnnnnnn.nnnnnnn>, <https://arxiv.org/abs/2410.21306v2>
5. Chalkidis, I., Fergadiotis, M., Androutsopoulos, I.: Legal-bert: The muppets straight out of law school. arXiv preprint arXiv:2010.02559 (2020)
6. Clavié, B., Alphonsus, M., Clavié, C., Mundi, J.: The unreasonable effectiveness of the baseline: Discussing svms in legal text classification (2021), <https://gitlab.com/jusmundi-group/public/Legal-svm-baselines>
7. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Burstein, J., Doran, C., Solorio, T. (eds.) Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). pp. 4171–4186. Association for Computational Linguistics, Minneapolis, Minnesota (Jun 2019). <https://doi.org/10.18653/v1/N19-1423>, <https://aclanthology.org/N19-1423/>
8. Falconí, L.G.: Transfer and ensemble learning models in breast mammogram pathology classification. Master’s thesis, Quito, 2020. (2020)
9. Howard, J., Ruder, S.: Universal language model fine-tuning for text classification (2018), <https://arxiv.org/abs/1801.06146>
10. Jouppi, N.P., Yoon, D.H., Kurian, G., Li, S., Patil, N., Laudon, J., Young, C., Patterson, D.: A domain-specific supercomputer for training deep neural networks. Communications of the ACM **63**(7), 67–78 (2020)
11. Kalia, A., Kumar, N., Namdev, N.: Classifying case facts and predicting legal decisions of the indian central information commission: a natural language processing approach. In: Advances in Deep Learning, Artificial Intelligence and Robotics, pp. 35–45. Springer (2022)
12. Kowsari, K., Meimandi, K.J., Heidarysafa, M., Mendu, S., Barnes, L., Brown, D.: Text classification algorithms: A survey (2019). <https://doi.org/10.3390/info10040150>, www.mdpi.com/journal/information
13. Liu, Z., Li, Z., Zhang, H.: Legal knowledge-enhanced language models for legal text processing. ICAIL (2022)
14. Minaee, S., Kalchbrenner, N., Cambria, E., Nikzad, N., Chenaghlu, M., Gao, J.: Deep learning-based text classification. ACM Computing Surveys (CSUR) **54** (4 2021). <https://doi.org/10.1145/3439726>, <https://dl.acm.org/doi/10.1145/3439726>

15. Mumcuoğlu, E., Öztürk, C.E., Ozaktas, H.M., Koç, A.: Natural language processing in law: Prediction of outcomes in the higher courts of turkey. *Information Processing & Management* **58**(5), 102684 (2021)
16. Murphy, K.P.: Probabilistic machine learning: an introduction. MIT press (2022)
17. Pathak, A.R., Pandey, M., Rautaray, S.: Application of deep learning for object detection. *Procedia Computer Science* **132**, 1706–1717 (2018). <https://doi.org/https://doi.org/10.1016/j.procs.2018.05.144>, <https://www.sciencedirect.com/science/article/pii/S1877050918308767>, international Conference on Computational Intelligence and Data Science
18. Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al.: Language models are unsupervised multitask learners. *OpenAI blog* **1**(8), 9 (2019)
19. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *ArXiv abs/1910.01108* (2019)
20. Savelka, J., Ashley, K.D.: Cross-domain generalization and knowledge transfer in transformers for legal text. *ICAIL* (2021)
21. Shaheen, Z., Wohlgenannt, G., Filtz, E.: Large scale legal text classification using transformer models (10 2020), <https://arxiv.org/abs/2010.12871v1>
22. Shaheen, Z., Wohlgenannt, G., Mourontsev, D.: Zero-shot cross-lingual transfer in legal domain using transformer models. *Proceedings - 2021 International Conference on Computational Science and Computational Intelligence, CSCI 2021* pp. 450–456 (11 2021). <https://doi.org/10.1109/CSCI54926.2021.00145>, <https://arxiv.org/abs/2111.14192v2>
23. Tunstall, L., von Werra, L., Wolf, T.: Natural language processing with transformers. " O'Reilly Media, Inc." (2022)
24. Vaswani, A., Shazeer, N., Parmar, N., et al.: Attention is all you need. *Advances in Neural Information Processing Systems* **30** (2017)
25. Vatsal, S., Meyers, A., Ortega, J.E.: Classification of us supreme court cases using bert-based techniques (2023), <https://huggingface.co/saibo/>
26. Wang, Y., Gao, J., Chen, J.: Deep learning algorithm for judicial judgment prediction based on bert. In: 2020 5th International Conference on Computing, Communication and Security (ICCCS). pp. 1–6. IEEE (2020)
27. Yan, G., Li, Y., Shen, S., Zhang, S., Liu, J.: Law article prediction based on deep learning. In: 2019 IEEE 19th International Conference on Software Quality, Reliability and Security Companion (QRS-C). pp. 281–284. IEEE (2019)
28. Yosinski, J., Clune, J., Bengio, Y., Lipson, H.: How transferable are features in deep neural networks? *Advances in neural information processing systems* **27** (2014)
29. Zhong, H., Xiao, C., Tu, C., Zhang, T., Liu, Z., Sun, M.: How does nlp benefit legal system: A summary of legal artificial intelligence. *Proceedings of the Annual Meeting of the Association for Computational Linguistics* pp. 5218–5230 (2020). <https://doi.org/10.18653/V1/2020.ACL-MAIN.466>, <https://aclanthology.org/2020.acl-main.466/>